

GEME: A Cognitive Primitive

Four Experiments in Emergent Concept Formation

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A personal summary, prepared for Douglas Hofstadter.

WHAT IS GEME

GEME (Genetic Economy Memory Entity) is a minimal system for emergent concept formation. It runs in about 200 lines of Python. No pretrained weights. No gradients. No external knowledge.

The idea: cognition is not statistical pattern extraction. It is not logical deduction. It is information competing for scarce memory.

Core mechanisms:

1. Vector encoding

Formulas (e.g., $\text{conn}(A,B)=\text{yes}$) become 27-dim frequency vectors.

Known symbols (swap, succ, conn, etc.) have dedicated dimensions.

Unknown symbols fall into generic "fn"/"const" slots.

2. Adaptive threshold

The system decides for itself what counts as "similar."

It collects distances during a bootstrap phase, then uses the median of successful merge distances as threshold.

No hardcoded constants.

3. Competitive memory

Fixed capacity. When a new formula arrives:

If vector is close enough to an existing frame --> merge
(centroid moves, weight increases by 1)

If not --> create a new frame

If capacity is exceeded --> evict the lowest-weight frame

4. Co-occurrence self-reference

A sliding window tracks which signatures appear close together.

Pairs that co-occur frequently generate "association frames" (--).

Associations that share an element generate "chain frames" (==).

This is self-reference: the system observes its own memory activity.

5. Induction (memory consolidation)

Periodically, half the frames are evicted. Unmerged frames decay

faster. Merged frames survive. This is the competitive selection that drives compression.

The entire system is about 200 lines of code.

EXPERIMENT 0: AUTONOMOUS CONCEPT CLASSIFICATION

Question: Can GEME separate different conceptual domains without being told which domain an input belongs to?

Setup:

Mix two formula types with different structural signatures:

Domain A (algebra/commutativity): $\text{sig} = \text{eq_swap_swap}$

Domain S (set/successor): $\text{sig} = \text{eq_succ}$

The system receives a random 50/50 stream. No domain labels.

Result:

Frame cluster purity: 100% -- every frame clearly belongs to A or S.

Within-domain associations: 67% of all associations are A-A or S-S.

Cross-domain associations: 33% (expected window co-occurrence noise).

Observation:

Conceptual domains are not externally labeled.

They are a natural product of information competition.

Different structural signatures lead to different attractor clusters.

The system divided its own world without being told how.

EXPERIMENT 1: COMMUTATIVITY AS EMERGENT CONCEPT

Question: Can GEME learn "commutativity" as a general concept from concrete instances -- $\text{swap}(1,2)=\text{swap}(2,1)$, $\text{swap}(3,5)=\text{swap}(5,3)$...?

Setup:

70% commutative: $\text{eq}(\text{fn}(\text{"swap"}, a, b), \text{fn}(\text{"swap"}, b, a))$

30% identity: $\text{eq}(\text{fn}(\text{"swap"}, a, b), \text{fn}(\text{"swap"}, a, b))$

Control: directional operation "dir" (non-commutative)

Result (verified across 5 random seeds, 42-46):

Memory: 5/16 frames -- 1 association (concept layer, w=47) + 4 instances

New commutative instances (unseen numbers): 5/5 recognized (eval=2)

Directional instances: 5/5 rejected (eval=3)

Frequency inversion test (70% -> 30%): concept persists

Observation:

The concept-instance separation is not programmed.

It emerges from the memory economy: specific pairs form instance frames;
their co-occurrence forms the concept frame at a higher level.

The relation ("commutativity") and the entities ("1", "2", "3"...)
automatically occupy different layers of the memory.

This is a computational echo of Kant's category of Relation
(Community/Reciprocity) -- not as an a priori given, but as a
structure that emerges from processing informational difference.

EXPERIMENT 2: SUCCESSOR AND THE GODEL WALL

Question: Can GEME form the "successor" concept? And if so, does it
encounter a boundary it cannot cross?

Setup:

All inputs: eq(fn("succ", X), Y), same sig = eq_succ

50% ordered chain: succ(s0)=s1, succ(s1)=s2, succ(s2)=s3...

50% scrambled pairs: succ(s5)=s2, succ(s8)=s3, succ(s9)=s4...

Both share the EXACT SAME structural signature eq_succ.

The only difference is semantic: ordered is meaningful, scrambled is not.

Result:

"succ" concept forms: association frame w=194, new instances recognized

Cross-function generalization fails correctly (next -> eval=3)

BUT: ordered and scrambled are INSEPARABLE at the signature level

This is not a data problem. It is not a parameter problem.

It is a structural limitation of the evaluation mechanism:

evaluate_sig() checks signature overlap but not vector content.

Two patterns with the same signature receive the same evaluation.

Observation (a computational encounter with Godel's Second Theorem):

Let G = GEME(memory, observe, evaluate, induction)

Let A = "succ(s5)=s2 is a meaningless random pairing"

sig(A) = eq_succ

evaluate_sig(eq_succ) -> S2 (belongs to concept)

But semantically, A IS meaningless. The system has no mechanism to detect this. Its evaluation function can only answer

"does the signature match?" -- not "is this instance meaningful?"

The derivation path depends on a premise the system cannot verify: its own evaluation function's consistency for distinguishing semantic validity from structural pattern matching.

--> BOUNDARY ENCOUNTER.

Godel's Second Theorem is not a philosophical metaphor here.

It is a running computational event in 270 lines of code.

EXPERIMENT 3: THE TARSKI WALL

Question: From pure connection assertions conn(A,B), can GEME learn to distinguish a closed triangle from an open chain?

Setup:

Test A (pure connection):

All inputs: eq(fn("conn", a, b), yes), sig = eq_conn

Closed triangle: conn(A,B), conn(B,C), conn(C,A) -- 3 assertions

Open chain: conn(A,B), conn(B,C) -- 2 assertions

Test B (control with different function names):

eq(fn("triangle", ABC), yes) vs eq(fn("chain", ABC), yes)

Result:

Test A: ALL conn assertions collapse into ONE concept (w=2616)

Closed and open are COMPLETELY indistinguishable.

Test B: With different function names, concepts separate normally.

This is Tarski's (1959) result -- "triangle is an axiom-level primitive, not derivable from connection primitives" -- running in a cognitive unit.

Not as a proof. As an event. 200 lines of code found the same wall that took human mathematics two thousand years to reach.

BREAKING THE WALL: THE WITTGENSTEIN TABLE

Layer 1 detects the burst: 3 conns together vs 2 conns together.

But it cannot conceptualize this difference (all sig = eq_conn).

Layer 2 receives the burst size encoded in function names:

burst3 (3 connections together) -> sig = eq_burst3

burst2 (2 connections together) -> sig = eq_burst2

Layer 2 CAN distinguish them -- they have different signatures.

The vocabulary ("burst3", "burst2") is not assigned by a human.

It comes from Layer 1's own co-occurrence statistics.

The system generated its own language.

What separated:

Layer 1: MEANING without LANGUAGE

It felt the difference (3 vs 2) but could not say it.

It hit the wall on the difference.

The wall -- Tarski's wall -- is the structural state of

"having meaning but no language to express it."

Layer 2: LANGUAGE without MEANING

It uses the symbols "burst3" and "burst2" without knowing

what they "refer to." It only knows they are different patterns.

The meaning of these words IS their use in the system --

which frames they co-occur with, how their weights change.

The bridge: Layer 1's co-occurrence statistics become

Layer 2's vocabulary.

This is Wittgenstein's language game (Philosophical Investigations, sec. 43) running in 200 lines of code.

"The meaning of a word is its use in the language" -- not as philosophy, but as a running computational process.

A FRAMEWORK DIALOGUE (not conclusions, observations)

KANT (Critique of Pure Reason, A51/B75):

"Thoughts without content are empty,

intuitions without concepts are blind."

Layer 1 has intuition (the felt difference of burst sizes)

but no concept (all collapsed under eq_conn). It is blind.

Layer 2 has concept (burst3, burst2 as distinguishable symbols)

but no intuition (it does not "see" triangles). It is empty.

Together -> knowledge. Separately -> neither.

The Kantian synthesis is not a philosophical ideal here.

It is an architectural condition for cognitive function.

GODEL (1931):

Experiment 2 is a computational instance of the Second Theorem.

The system cannot verify its own evaluation function's consistency.

It hits this wall in running code.

The wall is not a bug. It is a structural consequence of a self-referential system operating with finite expressive power.

What the First Theorem proves logically, GEME demonstrates computationally: every system with enough power to form concepts has concepts it cannot evaluate.

WITTGENSTEIN (Philosophical Investigations, 1953):

"For a large class of cases -- though not for all -- the meaning of a word is its use in the language."

In the wall-breaking experiment, "burst3" has no external referent.

Its meaning is entirely determined by its use in Layer 2:

which frames it merges with, what weight it accumulates, which other signatures it co-occurs with.

The language game is not a metaphor applied to the system.

It is the system's mode of operation.

TARSKI (1959, What is Elementary Geometry):

The triangle/chain inseparability in Experiment 3 is not a GEME limitation. It is the same boundary Tarski proved axiomatically: "triangle" is a primitive notion, not derivable from more basic geometric relations.

That GEME encountered this same wall -- independently, from a completely different direction (computational rather than logical) -- suggests the wall is not an artifact of any particular formalization. It is a structural fact about geometric knowledge.

HONEST BOUNDARIES

What GEME CAN do:

- * Form stable concept structures from information streams
- * Encounter and display its own boundaries (mathematical walls)
- * Through layering, form more abstract distinctions
- * Reproduce, in running code, findings from mathematical logic (Godel, Tarski) and philosophy of language (Wittgenstein)

What GEME CANNOT do:

- * Distinguish different semantics under the same structural signature within a single layer
- * Detect its own boundary encounters (needs an external observer)
- * Derive "triangle" from pure connection assertions (this is the Tarski wall -- not a bug)
- * Assign external reference to its self-generated vocabulary (the Wittgenstein table requires a second layer)

Concept is not understanding.

GEME forms concepts. It does not understand them.

This gap is not a defect to be fixed.

It is the central structural finding of this work.

OUTPUTS

Full experimental reports and design documents:

/docs/ -- all reports, English and Chinese

Core code (~200 lines):

src/gira/phase6/geme_v6.py

Reproducible experiment scripts (6):

scripts/_exp_*.py

Paper figures:

figures/fig1_frame_vectors.png

figures/fig2_weight_curve.png

"GEME is its first photograph."

I asked it: "Can you imagine the arrival of a GAI era?"

I kept the line. It was right.